

INDIVIDUAL TECHNICAL REPORT
Gamified Water Conservation Monitoring Portal

Environmental Problem: Water Wastage

SDG Focus: Sustainable Development Goal 6 – Clean Water and Sanitation

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1. Abstract

Water wastage is a significant environmental and resource-management problem caused by leakage, unnecessary consumption, inefficient practices, overflowing storage systems, and limited awareness of individual consumption patterns. Conventional awareness campaigns can communicate the importance of conservation, but they do not always provide continuous monitoring, feedback, or motivation for users to change their behaviour. This report proposes a Gamified Water Conservation Monitoring Portal, a web-based information technology solution that combines water-use monitoring, water-wastage reporting, conservation targets, and gamification.

The proposed portal allows users to enter periodic water-meter readings, view consumption trends, set conservation goals, report visible wastage, and participate in conservation activities and challenges. A gamification layer awards points, badges, levels, streaks, and leaderboard positions for verified or self-recorded conservation activities. Rather than claiming exact litres saved from individual behavioural actions, the system emphasizes measured changes in recorded consumption and transparent participation metrics. This makes the solution practical for a prototype without requiring dedicated hardware.

The proposed system supports Sustainable Development Goal 6, particularly the efficient use and sustainable management of water resources. Its environmental-management approach can also be aligned with the principles of ISO 14001 through monitoring, objectives, operational control, corrective action, and continual improvement. The report evaluates technical, economic, operational, and environmental feasibility and identifies expected benefits such as improved awareness, earlier identification of wastage, reduced consumption, stronger user participation, and data-driven conservation planning.

2. Introduction

Fresh water is a limited resource that is essential for human health, food production, ecosystems, industry, education, and economic activity. Although water is continuously renewed through the hydrological cycle, usable freshwater is unevenly distributed and is under pressure from population growth, urbanisation, climate variability, pollution, inefficient infrastructure, and increasing demand. Water conservation therefore requires not only the development of infrastructure but also responsible management and everyday behaviour.

Water wastage can occur in households, educational institutions, public facilities, commercial buildings, and industrial premises. Common examples include leaking taps and pipes, overflowing tanks, leaving taps running unnecessarily, excessive cleaning water, and poor monitoring of consumption. Small losses can become substantial when they occur continuously or across a large population. A major practical difficulty is that users may not know how much water they consume or how their current consumption compares with earlier periods.

Information technology can support conservation by converting consumption records into understandable dashboards, trends, alerts, goals, and reports. However, a monitoring system alone may not guarantee behaviour change. The proposed solution therefore introduces gamification. Gamification means applying game-inspired elements such as points, badges, challenges, levels, streaks, and leaderboards. In this project, these elements are used to motivate participation in water conservation while the underlying portal continues to perform the core function of monitoring water use and wastage.

The prototype is designed primarily for a college, residential community, or similar institution. In the initial version, water-meter readings can be entered manually. Future versions can integrate smart meters or IoT flow sensors for automated data collection. This staged approach keeps the prototype technically achievable while providing a pathway toward real-time monitoring.

3. Problem Statement

The central problem addressed by this project is the inefficient use and wastage of water due to inadequate monitoring, delayed identification of visible wastage, and insufficient motivation for users to adopt conservation practices.

Specific problems include:

- Users often have limited visibility into their daily, weekly, or monthly water consumption.
- Consumption data may be recorded but not converted into meaningful trends or comparisons.
- Minor leaks and other visible wastage may remain unreported for long periods.
- Conventional awareness messages may not provide continuous feedback or incentives.
- There may be no simple mechanism for setting personal or institutional water-conservation targets.
- Institutions may lack a centralized dashboard for monitoring participation and conservation performance.

- Exact water savings from individual behavioural actions are difficult to measure without dedicated flow sensors.

Therefore, the proposed system should provide a practical monitoring mechanism while encouraging sustained participation. It should distinguish between measured consumption data and estimated or self-reported conservation activities so that the system does not present uncertain estimates as exact environmental savings.

4. Sustainability Analysis

4.1 Relevant Sustainable Development Goals

The primary SDG associated with this project is SDG 6: Clean Water and Sanitation. The proposal is particularly relevant to Target 6.4, which focuses on substantially increasing water-use efficiency across sectors and ensuring sustainable withdrawals and supply of freshwater. A portal that makes consumption visible, encourages conservation, and identifies wastage supports the broader objective of improving water-use efficiency.

The project also has connections with SDG 12 (Responsible Consumption and Production), because it encourages responsible resource consumption, and SDG 11 (Sustainable Cities and Communities), where the portal could support efficient resource management in institutions and communities. SDG 13 (Climate Action) is also indirectly relevant because water supply, treatment, pumping, and distribution require energy; reducing unnecessary water use can therefore contribute to resource and energy efficiency.

4.2 ISO 14001 Relevance

ISO 14001 provides requirements and guidance for an environmental management system (EMS). The proposed portal is not itself an ISO 14001 certification system, but its functions can support several EMS principles. The monitoring dashboard provides environmental performance information; conservation goals support objectives and planning; wastage reports provide a mechanism for identifying issues; administrative follow-up supports corrective action; and historical data enables continual improvement.

- Environmental aspect identification: identify excessive water consumption, leaks, and other wastage sources.
- Objectives and targets: establish measurable reduction targets for a household, class, department, or building.
- Monitoring and measurement: collect meter readings and track consumption over time.

- Operational control: provide procedures and reminders for reporting and addressing wastage.
- Corrective action: track reported leaks or wastage until resolution.
- Continual improvement: compare historical performance and adjust conservation targets or activities.

4.3 LCA/EIA Applicability

A full Life Cycle Assessment (LCA) is not essential for the basic software prototype because the main intervention is a digital information system rather than a new physical product. Nevertheless, a simplified life-cycle perspective can be used to recognize the environmental impacts of user devices, servers, network operation, and eventual sensor hardware. If the project is expanded to include IoT meters, an LCA could consider manufacturing, transport, energy consumption, maintenance, and end-of-life disposal of the hardware.

A formal Environmental Impact Assessment (EIA) is also generally not required for a small software prototype. However, EIA-style thinking is useful when the system is deployed as part of a larger physical water infrastructure project. For example, the environmental and operational effects of installing sensors, modifying plumbing, or implementing large-scale water infrastructure would need to be assessed separately.

5. Proposed Engineering/IT Solution

The proposed individual solution is a Gamified Water Conservation Monitoring Portal. The unique aspect of this solution is the integration of a conventional water-monitoring portal with a structured gamification layer. The portal remains responsible for collecting, storing, visualizing, and reporting water-related data, while gamification encourages users to interact with the system and maintain conservation behaviour.

5.1 Objectives

- Provide a simple digital platform for monitoring recorded water consumption.
- Make consumption trends understandable through charts and comparisons.
- Allow users to set measurable conservation targets.
- Provide a structured mechanism for reporting water wastage.
- Encourage conservation through points, badges, challenges, levels, streaks and leaderboards.
- Provide administrators with aggregated data for institutional decision-making.
- Create a scalable foundation for future IoT-based automatic monitoring.

5.2 Major Modules and Features

Module	Function
User Management	Registration, login, profile, department/class information, and activity history.
Water Usage Monitoring	Periodic meter-reading entry, calculated consumption, usage history, trends, and comparisons.
Conservation Goals	Daily, weekly, or monthly targets with progress indicators.
Wastage Reporting	Reports for leaks, overflowing tanks, broken taps, and other visible wastage, with location and optional image.
Gamification Engine	Points, levels, badges, challenges, streaks, and leaderboard rankings.
Analytics	Charts for consumption, participation, conservation performance, and wastage reports.
Admin Dashboard	User management, report verification, challenge management, and institutional statistics.
Notifications	Goal reminders, challenge announcements, achievement notifications, and report-status updates.

5.3 Water Consumption Monitoring

In the prototype, users can enter two or more meter readings with dates. Consumption for a period is calculated as the difference between the current and previous valid reading. For example, if a previous reading is 12,450 units and the current reading is 12,780 units, the recorded consumption is 330 units, subject to the meter's measurement unit. The portal stores the reading, date, and user or location, then generates weekly and monthly visualizations.

The system can compare a current period with a previous period. For example, if consumption decreases from 1,000 L to 850 L over comparable periods, the portal can report a 15% reduction. This is preferable to assigning an exact litre-saving value to a

self-reported activity because the latter depends on assumptions about flow rate and duration.

5.4 Gamification Mechanism

The gamification system rewards actions and measurable progress. Example activities include completing a water-saving challenge, reporting a verified leak, meeting a conservation target, or maintaining a conservation streak. Points can be assigned by the administrator. Badges are unlocked when predefined conditions are met.

- Points: numerical score awarded for conservation activities and verified contributions.
- Levels: progression based on accumulated points, such as Water Beginner, Water Saver, Water Warrior, and Water Guardian.
- Badges: achievements for milestones such as first report, target achievement, or seven-day streak.
- Challenges: daily, weekly, and monthly conservation tasks.
- Streaks: rewards for maintaining participation over consecutive days or weeks.
- Leaderboard: ranking based on points or verified conservation participation.
- Recognition: optional digital certificates or institutional recognition for high-performing participants.

5.5 Wastage Reporting Workflow

1. A user identifies a visible water-wastage issue.
2. The user submits the location, issue type, description, date, and optional photograph.
3. The administrator reviews and verifies the report.
4. The issue is assigned a status such as Pending, Under Review, In Progress, or Resolved.
5. Once verified, the system can award points to the reporting user.
6. The portal updates institutional statistics and the user's activity history.

5.6 Proposed Technology Stack

- Frontend: HTML5, CSS3, and JavaScript for a responsive web interface.
- Backend: Python with Flask for routing, authentication, business logic, and API endpoints.
- Database: SQLite for the prototype; MySQL or PostgreSQL for a larger deployment.
- Charts: Chart.js for water-consumption and participation visualizations.
- Version control: Git and GitHub for source-code management.
- Deployment: A suitable cloud-hosting service can be used after local testing.

- Future hardware integration: ESP32/Arduino with water-flow sensors and a suitable communication method.

5.7 System Workflow

The overall workflow is: User registration → meter-reading/activity input → database storage → consumption calculation and analysis → dashboard visualisation → goal evaluation → points/badges/challenges → leaderboard update. Wastage reports follow a parallel workflow from submission to administrator verification and resolution.

6. Feasibility Analysis

6.1 Technical Feasibility

The proposed prototype is technically feasible because it can be developed using widely available web technologies. HTML, CSS, and JavaScript are sufficient for the interface, while Flask can implement server-side logic and SQLite can store prototype data. Chart.js provides ready-to-use visualization capabilities. The system does not require specialized hardware in its initial version.

6.2 Economic Feasibility

The software prototype has low initial cost because the core development tools are open-source or freely available for educational use. A basic deployment can be performed using low-cost or free-tier services, subject to their current terms. If IoT hardware is later introduced, additional costs will arise from flow sensors, microcontrollers, power supplies, installation, networking, maintenance, and calibration.

6.3 Operational Feasibility

The portal is designed for users with basic smartphone or computer skills. A simple dashboard, clearly labelled forms, and visual progress indicators can reduce the learning curve. Administrators can manage reports and challenges through a separate interface. Adoption will depend on data accuracy, user participation, and institutional support.

6.4 Environmental Feasibility

The digital solution has a relatively small direct environmental footprint compared with physical water infrastructure. Its main potential environmental benefit arises from influencing consumption and improving the identification of wastage. Environmental claims should be based on recorded data where possible, while self-reported activities should be treated as participation indicators.

6.5 Social and Ethical Considerations

Gamification should motivate rather than shame users. Leaderboards can be optional or anonymised, particularly where individual consumption data may be sensitive. User accounts should collect only necessary information, and access to administrative data should be restricted. The system should also avoid rewarding false reports or unverifiable claims without review.

7. Expected Environmental Benefits

- Improved visibility of water consumption and changes over time.
- Earlier reporting and resolution of visible leaks and other wastage.
- Greater awareness of individual and institutional water-use patterns.
- Increased participation in conservation activities through continuous feedback.
- Encouragement of measurable reductions in recorded consumption.
- Creation of historical data that can support future conservation planning.
- Potential reduction in unnecessary water demand and the energy associated with pumping and treatment.
- Scalability from a small college deployment to residential communities or other institutions.

The actual environmental benefit should be evaluated after deployment using baseline and post-implementation consumption data. Useful indicators include percentage change in water consumption, number of wastage reports, percentage of reports resolved, number of active participants, challenge completion rate, and percentage of users achieving conservation targets.

8. Justification of the Proposed Solution

The proposed solution is justified because water conservation is partly a behavioural and management challenge. A monitoring portal provides information, but information is more useful when users receive timely feedback and have clear goals. Gamification adds an engagement mechanism without changing the fundamental purpose of the portal.

The solution is also appropriate for an IT-focused individual project because it demonstrates several core computer-science concepts: user authentication, database design, CRUD operations, business rules, data visualisation, role-based access, ranking algorithms, and responsive web design. It can be implemented without expensive hardware while still leaving a clear path toward IoT integration.

A key design decision is to avoid treating estimated behavioural savings as exact measurements. The system therefore separates recorded meter consumption from self-

reported conservation actions. This improves the credibility of environmental reporting. At the same time, verified wastage reports can connect digital participation with real-world corrective action.

Compared with a portal that only displays water statistics, the proposed system aims to close the loop: monitor → inform → motivate → act → evaluate. This combination makes the solution distinct while remaining within the required Water Conservation Monitoring Portal concept.

9. Conclusion

Water wastage requires both effective monitoring and sustained conservation behaviour. The proposed Gamified Water Conservation Monitoring Portal addresses these needs through a web-based platform for recording water consumption, visualizing trends, setting conservation targets, reporting wastage, and encouraging participation through gamification.

The solution is feasible as a software prototype using common web-development technologies and does not require physical sensors in its first implementation. Points, badges, challenges, streaks, and leaderboards can motivate users, while administrator verification helps connect online activities to real-world water-wastage reduction. The project supports SDG 6 and can contribute to environmental-management practices consistent with ISO 14001 principles.

Future development can integrate smart water meters, IoT flow sensors, automated anomaly detection, mobile notifications, and more advanced analytics. With appropriate baseline and post-deployment measurements, the portal can also be evaluated quantitatively for its effect on water consumption and conservation behaviour.

10. References

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